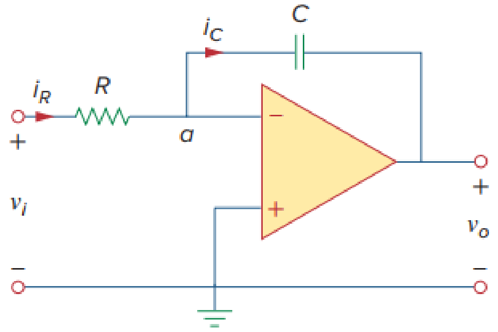
Problem

The integrator in Fig. 6.35(b) has R = 100 kΩ, C = 20 μF. Determine the output voltage when a dc voltage of 2.5 mV is applied at t = 0. Assume that the op amp is initially nulled.

Fig. 6.35(b): Replacing the feedback resistor in the inverting amplifier in (a) produces an integrator in (b).



Step-by-step solution

Step 1 of 3

Consider the following integrator circuit:

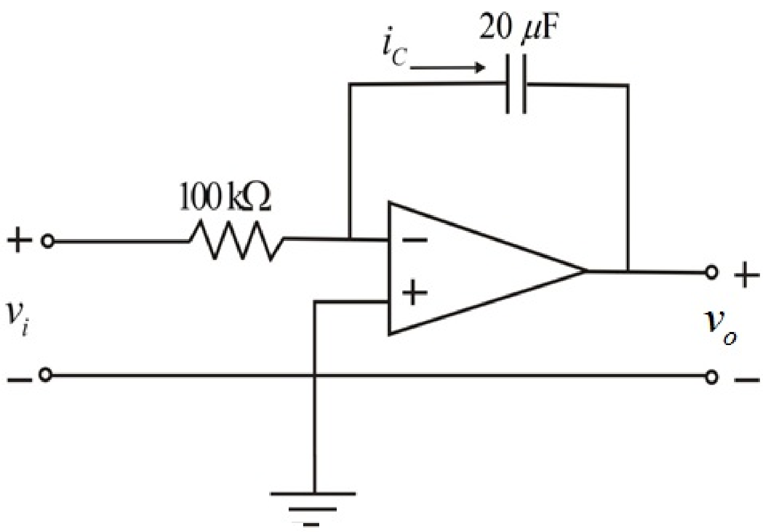


Figure 1

Step 2 of 3

The dc voltage, $v_i = 2.5$ mV at $t = 0$.

The current through the resistor is equal to that in the capacitor, that is, $i_R = i_C$ because both resistor and capacitor are in series.

$$\begin{aligned} i_R &= \frac{v_i}{R} \\ &= \frac{2.5\times10^{-3}}{100\times10^{3}} \\ &= 2.5\times10^{-8}\ \text{A} \end{aligned}$$

Step 3 of 3

Write the formula for capacitor current.

$$i_C = -C\frac{dv_o}{dt}$$

Here $i_R = i_C$.

Thus,

$$\frac{v_i}{R} = -C\frac{dv_o}{dt}$$

$$\begin{aligned} dv_o &= \left(-\frac{1}{RC}\right)v_i dt \\ &= \left(-\frac{1}{100\times10^3\times20\times10^{-6}}\right)\left(2.5\times10^{-3}\right)dt \\ &= -0.5\times2.5\times10^{-3}dt \\ &= -1.25\times10^{-3}dt \end{aligned}$$

Integrate the equation to get the output voltage.

$$\begin{aligned} v_o &= -1.25\times10^{-3}t\ \text{V} \\ &= -1.25t\ \text{mV} \end{aligned}$$

Thus, the output voltage, v_o of the integrator circuit is $\boxed{-1.25t\ \text{mV}}$.